Birla Institute of Technology and Science, Pilani
First Semester, 2015-16
CHE F212, Fluid Mechanics

Surprise Quiz 2(closed Book) 19/08/15
15 min, 5 points
NAME: **SOLUTION** ID: Section:

Consider the two-fluid manometer shown.
Calculate the applied pressure difference.
$SG_{ct} = 1.595$

Given: Two-fluid manometer as shown

$l = 10.2\cdot mm \quad SG_{ct} = 1.595$ (From Table A.1, App. A)

Find: Pressure difference

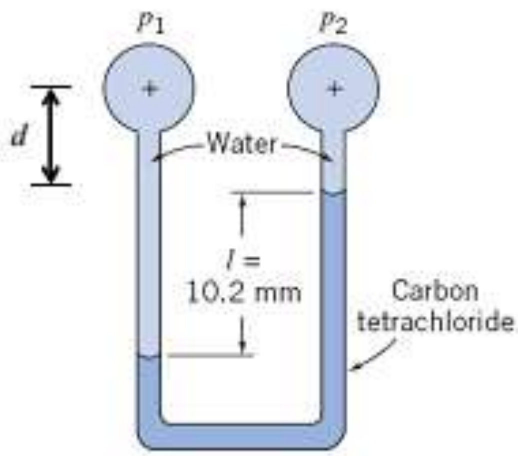
Solution: We will apply the hydrostatics equation.

Governing equations:

$$\frac{dp}{dh} = \rho \cdot g \qquad \text{(Hydrostatic Pressure - h is positive downwards)}$$

$$\rho = SG \cdot \rho_{water} \qquad \text{(Definition of Specific Gravity)}$$

Assumptions: (1) Static liquid
(2) Incompressible liquid



Starting at point 1 and progressing to point 2 we have:

$$p_1 + \rho_{water} \cdot g \cdot (d + l) - \rho_{ct} \cdot g \cdot l - \rho_{water} \cdot g \cdot d = p_2$$

Simplifying and solving for $p_2 - p_1$ we have:

$$\Delta p = p_2 - p_1 = \rho_{ct} \cdot g \cdot l - \rho_{water} \cdot g \cdot l = \left(SG_{ct} - 1\right) \cdot \rho_{water} \cdot g \cdot l$$

Substituting the known data:

$$\Delta p = (1.591 - 1) \times 1000 \cdot \frac{kg}{m^3} \times 9.81 \cdot \frac{m}{s^2} \times 10.2 \cdot mm \times \frac{m}{10^3 \cdot mm} \qquad \Delta p = 59.1\ Pa$$